SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Duration: 60 Minutes **Total Marks: 20**

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Code of Conduct

Gowriswaryd.b 192311232 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Gowriswarya

Criteria	Concept Identification (1.25)	Relationship Mapping (1.25)	Hierarchy and Organization(1)	Technical Relevance (1)	Visual Clarity(0.5)	TOTAL (5)
Weightage	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						

Signatureof the Faculty **TotalMarksAwarded:**

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Architecture: Include front-end, back-end, client devices, data centers, and distributed servers.
2. Concept Map 2: Characteristics of Cloud Computing: Connect on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service.
3. Concept Map 3: Deployment Models: Map public cloud, private cloud, hybrid cloud, and community cloud with usage scenarios.
4. Concept Map 4: Cloud Benefits vs Challenges: Show advantages (cost efficiency, scalability, flexibility) and limitations (security, downtime, vendor lock-in).

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

riteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
oncept entification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential conc are missing
lationship apping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships mostly incor
rarchy and ganization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organizatio
hnical evance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak align to topic
al Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly pre

Cloud Computing

Benifits

Cost efficiency
Reduces capital expenses pay-as-you-use model saves money

Scalability
resources can be scaled up or down quickly to meet demand.

Flexibility
Access resources anytime, anywhere
Supports innovation and agility

Cloud Deployment Models

PUBLIC CLOUD

Services delivered over the public internet and shared among multiple organizations

USAGE SCHEMAS

* web applications
* Startups
* Development & testing
* Big Data analytics

PRIVATE CLOUD

used exclusively by a single organizations over a private network

USAGE SCHEMAS

* Government
* Banking & finance
* Sensitive Data
* Internet Application

HYBRID CLOUD

combination of public and private clouds that remain unique entities but are bound together

USAGE SCHEMAS

* Data Backup
* Disaster recovery
* workload Bursting
* Phased Migration

COMMUNITY CLOUD

shared by several organizations with common concerns or requirements

USAGE SCHEMAS

* Healthcare organisations
* Educational institutions
* Compliance needs

users Can provision resources automatically without human interaction

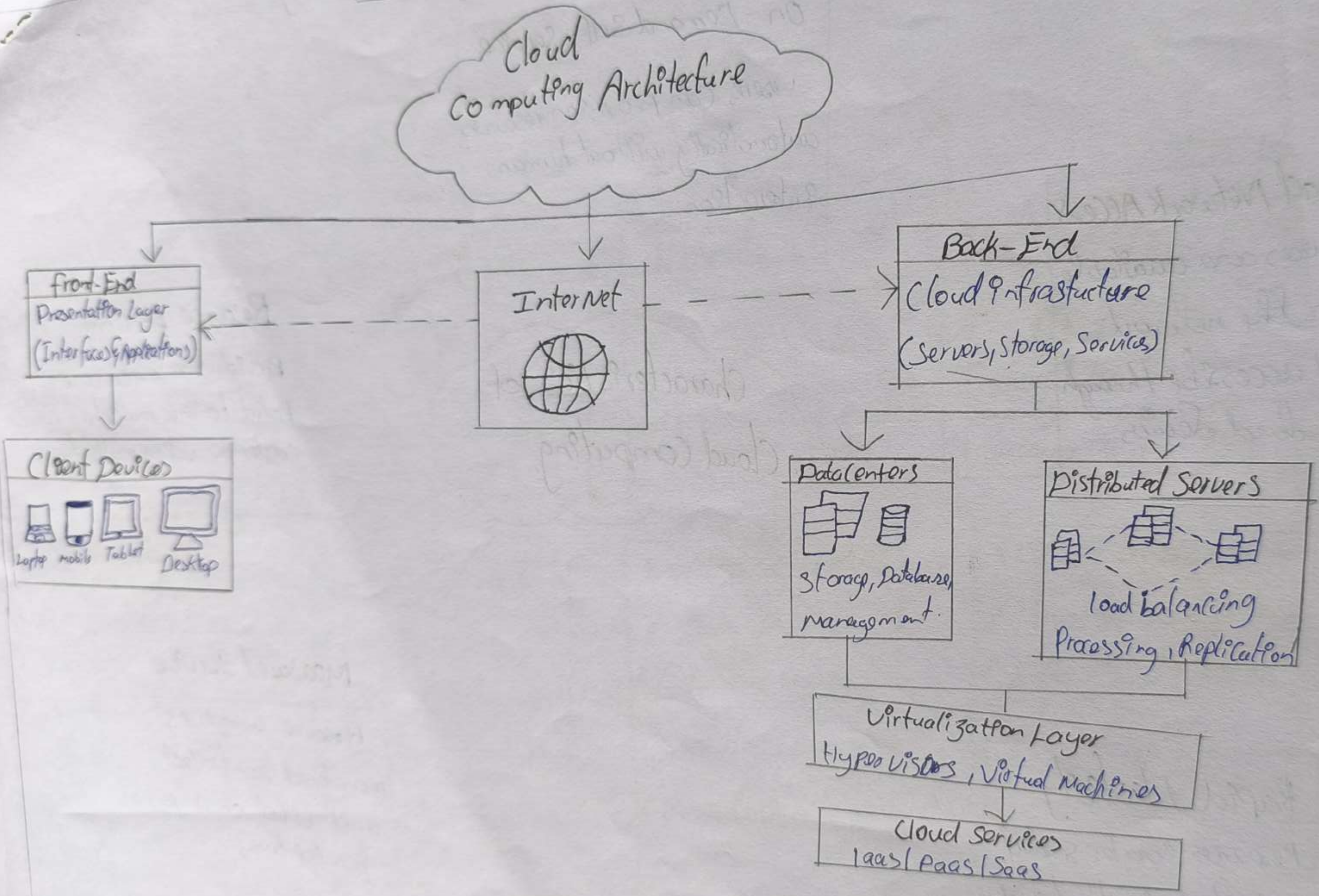
Characteristics of Cloud Computing

Broad Network Access

services are available over the network and accessible through standard devices.

Rapid Elasticity

Resources can be scaled up or down quickly based



Cloud Computing Architecture
Front-End
Presentation Layer
(Interfaces & Applications)
Client Devices
Laptop
mobile
Tablet
Desktop
Internet
Back-End
Cloud infrastucture
(Servers, Storage, Services)
Datacenters
storage, Database, management.
Distributed Servers
load balancing
Processing, Replication
Virtualization Layer
Hypervisors, Virtual Machines
Cloud Services
Iaas | Paas | Saas